

**MASTER OF PHARMACY (M. PHARM.):**

**DRUG REGULATORY AFFAIRS**

**(Two year post graduate degree course)**

**CREDIT BASED SEMESTER SYSTEM**

**SCHEME OF TEACHING, EXAMINATION AND SYLLABUS**

**M. Pharm. I Semester**

**w.e.f. ACADEMIC SESSION 2013-2014**



**RKDF University**

**Bhopal (MP) 462033, INDIA**



## M. Pharm : Drug Regulatory Affairs

### Scheme of Teaching and Examination

| M. Pharm I Semester Scheme |                                           |                            |    |         |    |                    |     |     |     |
|----------------------------|-------------------------------------------|----------------------------|----|---------|----|--------------------|-----|-----|-----|
| Subject Code               | Name of Subject                           | Teaching scheme (hrs/week) |    | Credits |    | Examination Scheme |     |     |     |
|                            |                                           | T                          | P  | T       | P  | T                  |     | P   |     |
|                            |                                           |                            |    |         |    | Ext                | Int | Ext | Int |
| MPMI101[T]                 | Modern Instrumental Methods               | 4                          | -  | 4       | -  | 70                 | 30  | -   | -   |
| MPMI101[P]                 | Modern Instrumental Methods               | -                          | 6  | -       | 3  | -                  | -   | 60  | 40  |
| MPDR102[T]                 | Drug Regulatory Affairs – I               | 4                          | -  | 4       | -  | 70                 | 30  | -   | -   |
| MPDR102[P]                 | Drug Regulatory Affairs – I               | -                          | 6  | -       | 3  | -                  | -   | 60  | 40  |
| MPPD103[T]                 | Product Development and Quality Assurance | 4                          | -  | 4       | -  | 70                 | 30  | -   | -   |
| MPPD103[P]                 | Product Development and Quality Assurance | -                          | 6  | -       | 3  | -                  | -   | 60  | 40  |
| MPDR104[T]                 | Total Quality Management systems          | 4                          | -  | 4       | -  | 70                 | 30  | -   | -   |
|                            |                                           | 16                         | 18 | 16      | 09 | 400                |     | 300 |     |
| Total                      |                                           | 34 hrs/week                |    | 25      |    | 700                |     |     |     |

T- Theory, P- Practical, Ext- External, Int- Internal

Theory: 1 Theory hour = 1 Credit

Practical: 2 Practical hour = 1 Credit

Internal assessment (Theory): Best one out of two sessional per semester.

Internal assessment (Practical) Based on day to day performance including attendance, *viva-voce* and laboratory record.



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|---------------------|-------------------------|---------------------|-----------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | First                       |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 60 Hrs [Theory]             |
| <b>Subject Code</b> | MPMI101[T]              | <b>Subject Name</b> | Modern Instrumental Methods |

## Syllabus

**Ultraviolet – Visible spectrometry:** Introduction, the nature of electronic excitation, the origin of UV band structure, principle of absorption spectroscopy, chromophore-  $\sigma \rightarrow \sigma^*$ ,  $\eta \rightarrow \sigma^*$ ,  $\pi \rightarrow \pi^*$  transitions, basics of instrumentation techniques, pharmaceutical applications. Woodward - Fisher rule for calculation of  $\lambda_{\text{max}}$ . Introduction to optical rotatory dispersion and circular dichroism. Derivative spectroscopy.

**Fourier Transform Infrared Spectroscopy:** Introduction, the infrared absorption process, the modes of vibrations, stretching and bending, bond properties and absorption trends, basics of instrumentation techniques, pharmaceutical applications. Interpretation of Infrared spectra.

**Nuclear Magnetic Resonance Spectroscopy:** High resolution  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectroscopy, theoretical calculation of chemical shift of various carbon atoms, techniques used for finding out types of carbon attached proton test (APT) distortion less energy polarization transfer (DEPT), homonuclear and heteronuclear correlation spectroscopy. Different 1D and 2D NMR correlation spectrophotometric techniques used as COSY, NOESY, HETCOR, INADEQUATE, HSBC, HMQC, etc. Use of this technique in determination of absolute configuration.

**Spectrometry of other important nuclei:** Introduction to  $^{15}\text{N}$ ,  $^{19}\text{F}$ ,  $^{31}\text{P}$ , basic concepts.

**Mass spectrometry:** Basic principle and theory involved, basics of instrumentation techniques, tandem mass spectrometry and its applications.

**HPLC:** Instrumentation covering detailed discussion of pumps, injector system, columns and detectors. Analytical method development, validation as per ICH guidelines and troubleshooting. Quantification methods used in HPLC. Ultra pressure liquid chromatography.

**HPTLC:** Basic instrumentation and its calibration. Analytical method development and its validation as per ICH guidelines. Quantification using HPTLC.

**Thermoanalytical techniques:** Differential Scanning Calorimetry (DSC), Thermogravimetry (TG), Thermo mechanical analysis (TMA): Principles instrumentation and applications (including interpretation of data) in pharmacy.



**Radio analytical techniques used in pharmaceuticals:** Isotopic dilution methods, Radioimmunoassay, ELISA etc.

**Microscopy:** Transmission electron microscopy (TEM), Scanning electron microscopy (SEM), cryomicroscopy, Atomic force microscopy (AFM), confocal microscopy.

**Statistics :** Collection and classification of experimental data and its statistical treatment, Probability- definition and laws of probability, regression and correlation, method of least squares, correlation coefficient and multiple regression, Test of significance and t-test, statistical quality control, process control, control chart, acceptance sampling plans.

**Suggested Readings/Books:**

1. Instrumental methods of chemical analysis by chatwal. K, Anand, 5th edition.
2. Organnic spectroscopy by Y.R.Sharma.
3. Text book of pharmaceutical analysis by S.Ravishankar.
4. Spectrometric identification of Organic Compounds, Robert. M. Silverstein et al, 7<sup>th</sup> Edition, 1981.
5. Principles of Instrumental Analysis by Donglas A. Skoog, James, J. Leary, 4th Edition.
6. Pharmaceutical Analysis – Modern Methods – Part A, Part B, James W. Munson –2001.
7. Practical Pharmaceutical Chemistry, Part two, A. H. Beckett & J. B. Stenlake – 4<sup>th</sup> Edition.
8. Instrumental Methods of Chemical Analysis – B. K. Sharma - 9th Edition.
9. Chromatography – P. D. Sethi, Dilip Charegaonkar, 2nd Edition.
10. Spectroscopy of Organic Compounds by P. S. Kalsi.
- 11.Organic Spectroscopy by William Kemp
- 12.Pharmaceutical drug analysis by Ashouthosh Kar
- 13.Quantitative analysis of Drugs and Formulations by Sethi



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|---------------------|-------------------------|---------------------|-----------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | First                       |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 120 Hrs [Practical]         |
| <b>Subject Code</b> | MPMI101[P]              | <b>Subject Name</b> | Modern Instrumental Methods |

## **Practicals:**

Practical exercises based on the topic mentioned in theory syllabus.



|                     |                         |                     |                           |
|---------------------|-------------------------|---------------------|---------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | First                     |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 60 Hrs [Theory]           |
| <b>Subject Code</b> | MPDR102[T]              | <b>Subject Name</b> | Drug Regulatory Affairs-I |

## Syllabus

### Pharmaceutical legislations in India:

Origin, development, scope, objectives and nature of Pharmaceutical legislation in India. History and ethics of profession of Pharmacy.

A study of regulatory aspects that affect drug product design, manufacture and distribution in India with special emphasis on the detailed study of the following Acts / Laws (with latest amendments)

- The Drugs and Cosmetics Act, 1940 and Rules there under.
- The Narcotics Drugs and Psychotropic Substances Act.
- Medicinal and Toilet Preparations (Excise Duties) Act, 1955.
- Drugs (Price Control) Order in force.
- Copy Right Act, Trade Mark Act, and Biodiversity Act,
- The Drugs and Magic Remedies (Objectionable Advertisements) Act, 1955.
- Prevention of Cruelty to Animals Act.
- Factory's Act.
- The Environmental Protection Act
- Consumer Protection Act
- Law of Torts
- Law of Contracts
- Monopolistic & Restrictive Trade Practices Act

**Globalization of drug industries:** Export import policy of drugs, WHO –certification, Trademarks and copyrights.

**Schedule M & U requirements-** Product development stage documentation, factory procedures – Standard Operating Procedures (SOPs) and Standard Test Procedures (STPs).

**Legal environment of business-** Need for government regulations; financial regulations, SEBI, BIFR, FEMA and others, contract act and sale of goods act, company act, corporate tax laws – direct and indirect.



**Pharmaceutical regulatory process in India:** Hierarchy and working flow of FDA in India, Roles of DCGA and CDSCO in drug control, Drug Control Authority and its documentation in the state.

**Suggested Readings/Books:**

1. Drugs and Cosmetics Act, 1940 and its rules, published by Ministry of health and family welfare, Government of India.
2. Pharmaceutical Jurisprudence, G.K. Jani.
3. The Pharmaceutical Regulatory Process, 2nd ed. – Ira R. Berry, Robert P. Martin
4. Medical Product Regulatory Affairs: Pharmaceutical , Diagnostics, Medical Devices – John J. Tobin and Gary Walsh.
5. FDA Regulatory Affairs: A Guide for Prescription Drugs, Medical Devices and Biologics, 2nd ed. – Douglas J. Pisano and David S. Mantus
6. Good Drug Regulatory Practices: a Regulatory Affairs Quality Manual (Good Drug Development Series) – Helene I. Dumitriu.
7. Pharmaceutical Patent Law – John R. Thomas.
8. <http://cdsco.nic.in>
9. Original laws published by Govt. of India.
10. Text Book of Forensic Pharmacy by Mithal B. M.; Vallabh Prakashan, New Delhi.
11. Laws of Drugs in India by Hussain.
12. Text Book of Forensic Pharmacy by Jain N. K.; Vallabh Prakashan, New Delhi.



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| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | First                     |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 120 Hrs [Practical]       |
| <b>Subject Code</b> | MPDR102[P]              | <b>Subject Name</b> | Drug Regulatory Affairs-I |

## **Practicals:**

Practical exercises based on the topic mentioned in theory syllabus.





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|---------------------|-------------------------|---------------------|-------------------------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | First                                     |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 60 Hrs [Theory]                           |
| <b>Subject Code</b> | MPPD103[T]              | <b>Subject Name</b> | Product development and quality assurance |

## Syllabus

**Preformulation Studies:** Introduction, goals of preformulation, physicochemical properties, criteria for selection of drug and excipients, compatibility tests. Partition coefficient, Solid state pharmaceuticals.

**Optimization:** Definition, need, advantages, Meaning of general terms involved in optimization process. Classification of optimization methods.

Brief description and importance of experimental design with special reference to designs adequate for large number of variables. Introduction of correlation & regression analysis & mathematical model, contour plots. Basic understanding with at least one example of following optimization techniques:-Simplex method, langarengian method, EVOP, Grid search method.

**Solubility and Solubilization:** Development of theoretical relationships of prognostic relevance, theory and techniques of solubilization of drugs including surfactant systems, co-solvents, solid state manipulations, complexation and chemical modifications, methods of solubility enhancement, factors influencing solubility.

**Designing of Pharmaceuticals:** Tablet formulation, Special tablets, preparation of components for compression, characterization of granulation, coating of tablets, evaluation of tablets, Equipments and processing problems in tablet.

**Liquids:** Formulation considerations of oral liquids, suspensions, emulsions, development of various products. Formulation consideration of parenteral, ophthalmic, depot products, large volume and small volume parenterals, environmental control and quality assurance in parenterals.

**Disperse systems:** Molecular dispersion, coarse dispersions- Physical stability of suspensions and emulsion, role of zeta potential in stability of coarse dispersions, theory of emulsification, micro and multiple emulsions, rheology of suspensions and emulsions. Drug kinetics in coarse disperse systems, drug diffusion in coarse dispersion systems.

Topical and rectal formulations and evaluation.



**Introduction to Controlled and Novel Drug Delivery Systems:** Sustained release dosage forms, Prodrug concept, nanoparticles, liposomes, resealed erythrocytes, transdermal and other novel drug delivery systems.

Pilot-plant and scale-up techniques.

In vitro and In-Vivo evaluation techniques, product formulation and cGMP

**Validation:** Concepts in validation, prospective, concurrent, retrospective validation & revalidation, validation of manufacturing, analytical and process validation and its application.

Basic concepts of quality control and quality assurance systems, source and control of quality variation of raw materials: containers, closures, personnel, environmental etc.

In-process quality tests, IPQC problems in Pharmaceutical industries. ICH guidelines.

Sampling Plans, Sampling and Characteristic Curves.

Master formula generation and maintenance, standard operating procedure (SOP) for different dosage forms.

### **Suggested Readings/Books:**

1. Lachmann, L., Lieberman, H.A. & Kanig, J.I.: The Theory and Practice of Industrial pharmacy. Lea and Fibiger, Philadelphia.
2. Banker, G.S. & Rhodes, C.T. : Modern Pharmaceutics, Marcel Dekker Inc. New York and Basel.
3. Turco, S. & King R.E. : Sterile Dosage Forms, Lea and Febiger, Philadelphia.
4. Bean, H.S., Backett, A.H. & Carless, J.E: Advances in Pharmaceutical Sciences, Academic Press, London and Newyork.
5. Jain, N.K.: Controlled and Novel Drug Delivery, CBS, Delhi.
6. Robinson, J.R. & Lee, V.H.L.: Controlled Drug Delivery, Marcel Dekker, New York and Basel.
7. Chien,Y.W.: Novel Drug Delivery Systems, Marcel Dekker, New York and Basel.
8. Jain N. K. Pharmaceutical Product Development, CBS Publisher, Delhi



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|---------------------|-------------------------|---------------------|-------------------------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | First                                     |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 120 Hrs [Practical]                       |
| <b>Subject Code</b> | MPPD103[P]              | <b>Subject Name</b> | Product development and quality assurance |

## **Practicals:**

Practical exercises based on the topic mentioned in theory syllabus.



|                     |                         |                     |                                  |
|---------------------|-------------------------|---------------------|----------------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | First                            |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 60 Hrs[Theory]                   |
| <b>Subject Code</b> | MPDR104[T]              | <b>Subject Name</b> | Total Quality Management systems |

## Syllabus

Concept of Quality, Total Quality Management. Quality by design, six sigma concept

Auditors, Auditing strategies and preparation of audits, Quality audit & audit check lists and Auditing of manufacturing facilities by International regulatory agencies. Conducting and Handling of internal/Domestic/International Regulatory Audits/ Customer specific audits /Pre approval inspections

Harmonization of regulatory requirements-The International Conference on Harmonization (ICH) process, guidelines to establish quality, safety and efficacy of drug substances and products. Study of ICH common technical documents, harmonization of pharmacopoeial standards The International Organization for Standardization (ISO) 9000 series of quality systems standards, ISO 14000

Quality evaluation and batch release: Change Control, Deviation-(planned and unplanned), Corrective Action and Preventive Action (CAPA), Handling of non-conformance, Vendor evaluation process, Out of specification (OOS), batch reconciliation and finished goods release, Market recalls & Market complaints.

Good Laboratory Practices (GLP): Scope of GLP, Quality assurance unit, Standard operating procedures (SOP), protocols for conduct of non clinical testing, control on animal house, report preparation and documentation. 6. National Accreditation Board for testing and Calibration Laboratory (NABL) certification and accreditation procedure

Stability testing: ICH and WHO guidelines, Photostability studies

Good Clinical Practices (GCP): International regulatory requirements for pharmaceutical development regarding clinical research practices. Current issues in GCP; standards for design, conduct, performance, monitoring, auditing, recording, analysis, and reporting of clinical trials. Schedule Y of Indian Drugs and Cosmetics Act 1940, Role of Regulatory affairs in Product development, Clinical phase, Preclinical Phase, Manufacturing phase and Marketing Phase. Indian Council of Medical Research (ICMR) Guidelines for Ethics in Biomedical Research.

### Suggested Readings/Books:



1. Quality planning and Analysis by JM Juran and FM Gryna, Tata McGrawHill- India.
2. Total Quality Management, Dale H. Besterfield, Pearson Education, 3rd Ed., 2003.
3. Total Quality Management, Principles, Implementation & Cases, Sharma D.D., Sultan Chand & Sons, New Delhi, 2000.
4. Fundamentals of Total Quality Management, Process Analysis and Improvement by Jens.J Daulgard, Kai Kriestensen and Gopal K.Kanji. Taylor and Francis
5. Total Quality Management, Organization, and Strategy, James R. Evans, Thomson, 4th Ed., 2007.
6. Quality Control, Besterfield, D.H., Pearson, 7th Ed., 2004. 7. Implementing ISO 14000: a practical, comprehensive guide to the ISO 14000 environmental management standards, Authors: Tom Tibor, Ira Feldman , Editors: Tom Tibor, Ira Feldman, Irwin Professional Pub., 1997.
8. Establishing A cGMP Lab; Audit System- A practical guide, David M.Bleisner, Wiley Interscience.
9. The manager's guide to ISO 9000, Kenneth L. Arnold, Free Press, 1994.
- 10.How To Practice GLP, Good Laboratory Practice, Sharma PP, Vandana Publications
- 11.GLP Essentials: A Concise Guide to Good Laboratory Practice, Second Edition, Milton A. Anderson, Informa Healthcare.
12. GLP Quality Audit Manual, Milton A. Anderson, Third Edition, Informa Healthcare.
- 13.Laboratory Auditing for Quality and Regulatory Compliance, by Donald C.Singer, Stefan and Stedan, Drugs and Pharmaceutical Sciences, Vol.150
14. Handbook of Stability Testing in Pharmaceutical Development: Regulations, Methodologies, and Best Practices, Huynh-Ba, Kim, Springer.
15. International Stability Testing, Mazzo J. Mazzo, David J. Mazzo, Informa Healthcare
- Pharmaceutical Stability Testing To Support Global Markets (biotechnology: Pharmaceutical Aspects), Kim Huynh-ba, Springer.
16. Good Laboratory Practice Regulations, Third Edition, Revised and Expanded Edited by Sandy Weinberg
17. Handbook of Stability Testing in Pharmaceutical Development: Regulations, Methodologies, and Best Practices, Kim Huynh-ba, Springer.
18. Good Clinical Practice: Standard Operating Procedures for Clinical Researchers, Wiley.



19. Laboratory Auditing for quality and regulatory compliance, Donald C. Singer, Taylor and Francis.
20. Current Good Manufacturing Practices, MA Potdhar, BS Publications.



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**MASTER OF PHARMACY (M. PHARM.):**

**DRUG REGULATORY AFFAIRS**

**(Two year post graduate degree course)**

**CREDIT BASED SEMESTER SYSTEM**

**SCHEME OF TEACHING, EXAMINATION AND SYLLABUS**

**M. Pharm. II Semester**

**w.e.f. ACADEMIC SESSION 2013-2014**



**RKDF University**

**Bhopal (MP) 462033, INDIA**



## M. Pharm : Drug Regulatory Affairs

### Scheme of Teaching and Examination

| M. Pharm II Semester Scheme |                                       |                 |    |         |    |                    |     |     |     |
|-----------------------------|---------------------------------------|-----------------|----|---------|----|--------------------|-----|-----|-----|
| Subject Code                | Name of Subject                       | Teaching scheme |    | Credits |    | Examination Scheme |     |     |     |
|                             |                                       | T               | P  | T       | P  | T                  |     | P   |     |
|                             |                                       |                 |    |         |    | Ext                | Int | Ext | Int |
| MPDR201[T]                  | Drug Regulatory Affairs – II          | 4               | -  | 4       | -  | 70                 | 30  | -   | -   |
| MPDR202[T]                  | Drug Regulatory Affairs – III         | 4               | -  | 4       | -  | 70                 | 30  | -   | -   |
| MPDR203[T]                  | International Regulatory Requirements | 4               | -  | 4       | -  | 70                 | 30  | -   | -   |
| MPDR204[T]                  | Intellectual Property Rights          | 4               | -  | 4       | -  | 70                 | 30  | -   | -   |
| MPDR205[P]                  | DRA Practical II                      | -               | 18 | -       | 09 | -                  | -   | 60  | 40  |
|                             |                                       | 16              | 18 | 16      | 09 | 400                |     | 100 |     |
| Total                       |                                       | 34 hrs/week     |    | 25      |    | 500                |     |     |     |

T- Theory, P- Practical, Ext- External, Int- Internal

Theory: 1 Theory hour = 1 Credit

Practical: 2 Practical hour = 1 Credit

Internal assessment (Theory): Best one out of two sessional per semester.

Internal assessment (Practical) Based on day to day performance including attendance, *viva-voce* and laboratory record.





|                     |                         |                     |                            |
|---------------------|-------------------------|---------------------|----------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | Second                     |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 60 Hrs [Theory]            |
| <b>Subject Code</b> | MPDR201[T]              | <b>Subject Name</b> | Drug Regulatory Affairs-II |

## Syllabus

**Introduction to US FDA:** A detailed study of Federal Food, Drugs and Cosmetics Act of USA, restricted to human drugs, cosmetics and biotechnological products, with special emphasis on:

- Organization and functions of FDA, including historical developments.
- General definitions.
- Adulterated & misbranded drugs/cosmetics/biotechnological products.
- OTC drugs, Orphan drugs, Orange Book and Fast Track Products.
- General penalties as applicable to drugs, cosmetics and biotechnological products.

**US FDA-I:** A detailed study of Federal Food, Drugs and Cosmetics Act of USA, restricted to human drugs, cosmetics and biotechnological products, with special emphasis on:

- General drug approval process.
- Investigational New Drug application.
- New Drug Application and BLA.
- ANDA.
- SNDA, SUPAC and BACPAC.
- Post marketing surveillance.

**US FDA- II:** A detailed study of Federal Food, Drugs and Cosmetics Act of USA, restricted to human drugs, cosmetics and biotechnological products, with special emphasis on :

- Labelling and advertising requirements for drugs, cosmetics and biotechnological products.
- Introduction to environmental protection laws, as applicable to drugs, cosmetics and biotechnological products, including EPA and OSHA.
- Common Technical Document and Drug Master Files.
- Factory Inspection.

**International Conference on Harmonisation** of Technical Requirements For Registration of Pharmaceuticals For Human Use: History, structure and process for harmonisation.

**ICH guidelines on quality:** Stability testing of new drug substances and products stability testing : photostability testing of new drug substances and products, stability testing for new



dosage forms, bracketing and matrixing designs for stability testing of new drug substances and products, evaluation of stability data, impurities in new drug substances, impurities in new drug products, impurities: guideline for residual solvents,

**ICH guidelines on efficacy:** ICH guidelines on clinical trial and Good Clinical Practice

**ICH Guidelines on safety:** Carcinogenicity studies - need for carcinogenicity, studies of pharmaceuticals and testing for carcinogenicity of pharmaceuticals. genotoxicity: a standard battery for genotoxicity, testing of pharmaceuticals.

**Suggested Readings/Books:**

1. Medical Product Regulatory Affairs: Pharmaceuticals, Diagnostics, Medical Devices By John J. Tobin and Gary Walsh
2. FDA Regulatory Affairs: A Guide for Prescription Drugs, Medical Devices, and Biologics, Second Edition by Douglas J. Pisano and David S. Mantus
3. Good Drug Regulatory Practices: A Regulatory Affairs Quality Manual (Good Drug Development Series, Vol 1) by Helene I. Dumitriu
4. Encyclopedia of Pharmaceutical Technology, Jasmes Swarbrick and James C. Boylan, Marcel Dekker Inc., New York.
5. Guidelines for Developing National Drug Policies; WHO Publications, 1998.
6. Export Marketing by Cherian and Parab; Himalaya Publishing House, Delhi



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| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | Second                      |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 60 Hrs [Theory]             |
| <b>Subject Code</b> | MPDR202[T]              | <b>Subject Name</b> | Drug Regulatory Affairs-III |

## Syllabus

### **Drug regulatory authorities in European Union (EU) with special reference to EMA and UKMHRA:**

- Introduction, Organization and General Guidelines.
- Regulatory consideration for pre-clinical testing and clinical testing in EU.
- Registration application for marketing approval (IND, NDA, ANDA) in EU, Drug Master Files in EU.
- Regulatory considerations for manufacturing, packaging and labelling of pharmaceuticals in EU.

### **The WHO Guidelines:**

- The WHO Guidelines and their relevance in international registration.
- The WHO certification scheme on the quality of pharmaceutical product moving in international commerce.
- Introduction to Pharmacovigilance.
- Plant Layout as per WHO Guidelines.

**Guidelines for Rest of World (ROW):** African countries (MCC), Australia (TGA), Brazil (ANVISA), Japan, CIS countries, Gulf countries etc.

**Auditing of manufacturing facilities** by International regulatory agencies. The ISO 9000 series of quality systems standards.

**Status of pharmaceutical industry** with special reference to post GATT scenario. Project planning and implementation.

**Development of orphan drug:** Introduction, Designation process, tax credit, PDUFA (Prescription drug user fee act) and Orphan products development, Clinical trial design for rare disease treatment.

### **Suggested Readings/Books:**

1. Guidelines for Developing National Drug Policies; WHO Publications, 1998.
2. Export Marketing by Cherian and Parab; Himalaya Publishing House, Delhi



3. The Pharmaceutical Regulatory Process, 2nd ed. – Ira R. Berry, Robert P. Martin Medical Product Regulatory Affairs: Pharmaceutical, Diagnostics, Medical Devices John J. Tobin and Gary Walsh
4. Good Drug Regulatory Practices: a Regulatory Affairs Quality Manual (Good Drug Development Series) – Helene I. Dumitriu.
5. OPPI-Quality Assurance.
6. Quality Assurance of Pharmaceuticals–A Compendium of Guidelines and Related Materials, Vol.–1; WHO Publications.
7. Good Manufacturing Practices for Pharmaceuticals-A Plan for Total Quality Control by Sidney H. Willing & James R Stoker. (Drugs & Pharm. Sciences) Vol. 78; Marcel Dekker Inc.
8. Original laws published by Govt. of India.
9. Text Book of Forensic Pharmacy by Mithal B. M.; Vallabh Prakashan, New Delhi.
10. Laws of Drugs in India by Hussain.
11. Guidelines for Developing National Drug Policies; WHO Publications, 1998.
12. Relevant articles from journals.



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|---------------------|-------------------------|---------------------|---------------------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | Second                                |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 60 Hrs [Theory]                       |
| <b>Subject Code</b> | MPDR203[T]              | <b>Subject Name</b> | International Regulatory Requirements |

## Syllabus

Generic drug product development: introduction, quality control and quality assurance (QC & QA), Hatch-Waxman update, Drug product performance- *in vitro*, ANDA regulatory approval process, bioequivalence and drug product assessment- *in vivo*, scale up post approval changes, post marketing surveillance, outsourcing bioavailability and bioequivalence studies to contract research organizations. Formats for marketing authorization submission to US, EU, Asia-PAC (includes countries of East Asia, southeast Asia, Australasia, Oceania) etc., data privacy protection, pharmaceutical labeling, advertising and promotion, risk management in regulatory affairs

Regulatory requirements for product approvals: Active pharmaceutical ingredients, biologics, novel therapies, special categories [ Over the counter (OTCS), herbal medicines and Homeopathics] obtaining New Drug Application (NDA), Abbreviated New Drug Application (ANDA) for generic drugs, ways and means of US Registration for foreign drugs, Chemistry, Manufacturing and controls (CMC), Post approval Regulatory affairs, Regulation for combination products (Controlled release systems), medical device, Environmental concerns and regulations 21 Code of Federal Regulations (CFR) Part 11 and LIMS (Laboratory information Management System).

FDA Approvable indications and other considerations: Data procession for Global submission, Text and Tabular exposition- Common Technical Document (CTD)/ electronic Common Technical Document (eCTD) Format, working with contract Research Organization (CRO), Industry and FDA Liaison, Role of European Commission Competent Authorities and Notified Bodies and USFDA Authorities

Nonclinical drug development: Global submission of Investigational New Drug application (IND), New Drug application (NDA), Abbreviated New Drug Application (ANDA), Investigation medicinal product Dossier (IMPD) & Investigator Brochure (IB), New product Applications for Global Pharmaceutical Product approvals, US NDA vs Global CTD Formats, ANDA & Supplemental Abbreviated New Drug Application (SNDA), CTD and eCTD for



registration of pharmaceuticals for Human use, combination products (Controlled release systems).

Clinical trials: Developing clinical trial protocols, Institutional Review Board/Independent Ethics committee-formation and working procedures, Informed consent-process and procedures, HIPAA- A new requirement to clinical study process, Code of Federal Regulations (CFR)/ International Conference on Harmonization (ICH)/EU GCP obligations of Investigators, sponsors & Monitors, Importance of Quality Assurance in clinical trials, Managing and Monitoring clinical trials, European clinical trials (CT) directives implementation and update., Pharmacovigilance-safety monitoring in clinical trials.

### **Suggested Readings/Books:**

1. Generic Drug Product Development, Solid Oral Dosage forms, Leon Shargel and Isader Kaufer, Marcel Dekker series, Vol.143
2. The Pharmaceutical Regulatory Process, Edited by Ira R. Berry Marcel Dekker Series, Vol.144
3. The Pharmaceutical Regulatory Process, Second Edition Edited by Ira R. Berry and Robert P. Martin, Drugs and the Pharmaceutical Sciences, Vol.185 Informa Health care Publishers.
4. New Drug Approval Process: Accelerating Global Registrations By Richard A Guarino, MD, 5th edition, Drugs and the Pharmaceutical Sciences, Vol.190.
5. Guidebook for drug regulatory submissions / Sandy Weinberg. By John Wiley & Sons. Inc.
6. FDA regulatory affairs : a guide for prescription drugs, medical devices, and biologics /edited by Douglas J. Pisano, David Mantus.
7. Clinical Trials and Human Research: A Practical Guide to Regulatory Compliance By Fay A. Rozovsky and Rodney K. Adams
8. HIPAA and Human Subjects Research: A Question and Answer Reference Guide By Mark Barnes, JD, LL.M and Jennifer Kulynych, JD, PhD
9. Principles and Practices of Clinical Research, Second Edition Edited by John I. Gallin and Frederick P. Ognibene
10. Drugs: From Discovery to Approval, Second Edition By Rick Ng
11. New Drug Development: A Regulatory Overview, Eighth Edition By Mark Mathieu
12. Pharmaceutical Risk Management By Jeffrey E. Fetterman, Wayne L. Pines and Gary H. Slatko



13. Preparation and Maintenance of the IND Application in eCTD Format By William K. Sietsema
14. Medical Device Development: A Regulatory Overview By Jonathan S. Kahan
15. Medical Product Regulatory Affairs: Pharmaceuticals, Diagnostics, Medical Devices By John J. Tobin and Gary Walsh



|                     |                         |                     |                              |
|---------------------|-------------------------|---------------------|------------------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | Second                       |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 60 Hrs [Theory]              |
| <b>Subject Code</b> | MPDR204[T]              | <b>Subject Name</b> | Intellectual Property Rights |

## Syllabus

**Intellectual property concepts:** · Concept of property, conventional property Vs Intellectual Property. Basic aspect of the 8 different IPR mechanism Viz. patents, copyright, trademark, industrial design, layout design of integrated circuits, geographical indicators, plant varieties & trade secrets. WTO, TRIPS and TRIMS.

**IPR and Pharmaceutical Research:** · Benefits of IPRs to improve the quality of research work. Strategies for avoiding research duplications, infringements

**Practical aspect of patenting:** · Indian patent act and its recent amendment with respect to following aspect

- Patentable and non-patentable inventions.
- Essential criteria for filing a patent.
- Filing a patent in India and abroad
- Drafting of patent application
- Patenting: Regional routes & national routes.
- Introduction to World Intellectual Property Organization. (WIPO)
- Commercialization of patent: Need for commercialization of research and role of IPRs in research commercialization.
- Bayh Dole Act and Technology Transfer.
- Benefit/Disadvantages of patenting to the society
- Patent Exemptions.

### IPR related treaties:

- Patent co-operative treaty
- Budapest treaty

Sewage disposal and pollution control from pharmaceutical Industry: Categorization of pharmaceutical industry as per EPA, Solid waste management of the expiry and rejected materials. Biomedical waste (Management and Handling) Rules, 1998

Basic concepts of process validation





Planning & managing a validation program including change control, scale-up and post-approval changes (SUPAC), pre approval inspections (PAI) & technology transfer issues, validation of water & thermal systems, including heat ventilation and air conditioning (HVAC), facilities & cleaning validation, process validation of active pharmaceutical ingredients (APIs) and finished products, validation of sterile and non-sterile facility.

Validation of analytical methods, automated systems, international conference on harmonization (ICH) and World Health Organization (WHO) Guidelines for calibration of equipments, validation of process: mixing, granulation, drying, compression, filtration, filling.

### **Case Study**

#### **Suggested Readings/Books:**

1. IPR Handbook for Pharma Students and Researchers, Parikshit Bansal, Pharma Book Syndicate, Hyderabad.
2. Patents, N. R. Subbaram, Pharma Book Syndicate, Hyderabad.
3. Relevant articles from journals.
4. [www.pat2pdf.org](http://www.pat2pdf.org)
5. [www.patentstorm.us](http://www.patentstorm.us)
6. [www.freepatentsonline.com](http://www.freepatentsonline.com)
7. <http://www.wipo.int/pctdb/en/>
8. [www.espacenet.com](http://www.espacenet.com)
9. Pharmaceutical Process Validation, 3rd Edition, Edited by Robert Nash and Alfred Wachter, Marcel Dekker
10. Good Manufacturing Practices for Pharmaceuticals: A Plan for Total Quality Control from Manufacturer to Consumer, Sidney J. Willig, Marcel Dekker, 5<sup>th</sup> Ed.
11. Validation of Pharmaceutical Processes: Sterile Products, Frederick J. Carlton (Ed.) and James Agalloco (Ed.), Marcel Dekker, 2nd Ed.
12. Validation Standard Operating Procedures: A Step by Step Guide for Achieving Compliance in the Pharmaceutical, Medical Device, and Biotech Industries, Syed Imtiaz Haider
13. Pharmaceutical Equipment Validation: The Ultimate Qualification Handbook, Phillip A. Cloud, Interpharm Press
14. Pharmaceutical Quality Assurance by Manohar A. Potdhar, 2nd edition, Nirali Prakashan.



|                     |                         |                     |                     |
|---------------------|-------------------------|---------------------|---------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | Second              |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 120 Hrs [Practical] |
| <b>Subject Code</b> | MPDR205[P]              | <b>Subject Name</b> | DRA Practical       |

## **Practicals:**

Practical exercises based on the topics mentioned in theory syllabus of second semester.



**RKDF University**

**MASTER OF PHARMACY (M. PHARM.):**

**DRUG REGULATORY AFFAIRS**

**(Two year post graduate degree course)**

**CREDIT BASED SEMESTER SYSTEM**

**SCHEME OF TEACHING, EXAMINATION AND SYLLABUS**

**M. Pharm. III Semester**

**w.e.f. ACADEMIC SESSION 2013-2014**



**RKDF University**

**Bhopal (MP) 462033, INDIA**



## M. Pharm : Drug Regulatory Affairs

### Scheme of Teaching and Examination

| M. Pharm III Semester |                  |                            |    |         |    |                    |     |     |
|-----------------------|------------------|----------------------------|----|---------|----|--------------------|-----|-----|
| Subject Code          | Name of Subject  | Teaching Scheme (hrs/week) |    | Credits |    | Examination Scheme |     |     |
|                       |                  | T                          | P  | T       | P  | T                  | P   |     |
|                       |                  |                            |    |         |    |                    | Ext | Int |
| MPDRDIS-I             | Dissertation - I | -                          | 36 | -       | 18 | -                  | 180 | 120 |
| Total                 |                  | 36 hrs/week                |    | 18      |    | 300                |     |     |

T- Theory, P- Practical, Ext- External, Int- Internal

Practical: 2 Practical hour = 1 Credit

Internal assessment (Practical) Based on dissertation work and *viva-voce*.



|                     |                         |                     |                  |
|---------------------|-------------------------|---------------------|------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | Third            |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 36 Hrs/Week      |
| <b>Subject Code</b> | MPDRDIS-I               | <b>Subject Name</b> | Dissertation - I |

Research work during third semester.

The examination shall be based on dissertation - I submitted at the end of third semester and presentation in open seminar.



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**CREDIT BASED SEMESTER SYSTEM**

**SCHEME OF TEACHING, EXAMINATION AND SYLLABUS**

**M. Pharm. IV Semester**

**w.e.f. ACADEMIC SESSION 2013-2014**



**RKDF University**

**Bhopal (MP) 462033, INDIA**



**M. Pharm : Drug Regulatory Affairs**

**Scheme of Teaching and Examination**

| <b>M. Pharm IV Semester</b> |                          |                                       |           |                |           |                           |            |            |
|-----------------------------|--------------------------|---------------------------------------|-----------|----------------|-----------|---------------------------|------------|------------|
| <b>Subject Code</b>         | <b>Name of Subject</b>   | <b>Teaching Scheme<br/>(hrs/week)</b> |           | <b>Credits</b> |           | <b>Examination Scheme</b> |            |            |
|                             |                          | <b>T</b>                              | <b>P</b>  | <b>T</b>       | <b>P</b>  | <b>T</b>                  | <b>P</b>   |            |
|                             |                          |                                       |           |                |           |                           | <b>Ext</b> | <b>Int</b> |
| <b>MPDRDIS-II</b>           | <b>Dissertation - II</b> | <b>-</b>                              | <b>36</b> | <b>-</b>       | <b>18</b> | <b>-</b>                  | <b>180</b> | <b>120</b> |
| <b>Total</b>                |                          | <b>36 hrs/week</b>                    |           | <b>18</b>      |           | <b>300</b>                |            |            |

T- Theory, P- Practical, Ext- External, Int- Internal

Practical: 2 Practical hour = 1 Credit

Internal assessment (Practical) Based on dissertation work and viva-voce.



|                     |                         |                     |                   |
|---------------------|-------------------------|---------------------|-------------------|
| <b>Course</b>       | M. Pharm                | <b>Semester</b>     | Fourth            |
| <b>Branch</b>       | Drug Regulatory Affairs | <b>Duration</b>     | 36 Hrs/Week       |
| <b>Subject Code</b> | MPDRDIS-II              | <b>Subject Name</b> | Dissertation - II |

Research work during third and fourth semester.

The examination shall be based on dissertation - II submitted at the end of Fourth semester and presentation in open seminar.